Nmap - Host Discovery

Aim: To study Nmap host discovery techniques and identify live hosts in a network using ARP, ICMP, TCP and UDP scans.

Task 1 - Introduction to Nmap

What is Nmap

Nmap (Network Mapper) is a free and open-source network scanning tool used to discover hosts, networks and services.

What created Nmap?

* Nmap was created by Gordon Lyon (Fyodor)

What are the main purposes of Nmap?

1. To discover live hosts
2. To identify running services
3. To scan networks and ports
4. To perform network reconnaissance

Task 2 - Subnetworks

How many devices can see the ARP Request sent from computer1? 4.

Did computer6 receive the ARP Request?

Nay

How many devices can see the ARP Request sent from computer_4? 4

Did computer6 reply to the ARP Request?

Yea

ARP (Address Resolution Protocol) is used to find the MAC address of a device on the same subnet.

ARP requests cannot cross a router.

Task 3 - Host Discovery using TCP/IP

What type of packet did computer1 send before Ping?

* ARP Request

what type of packet did computer1 receive before sending the Ping? ARP Response

How many computers responded to the Ping Request?

1

What was the first device that responded to the first ARP Request? Router

What was the first device that responded to the second ARP Request? Computer5

Did another Ping Request require new ARP Requests?

* Nay

Task 4 - Enumerating Targets

What is the first IP address Nmap would scan for 10.10.12.13/29

* 10.10.12.8

How many IP addresses will Nmap scan for 10.10.0-255.101-125?

* 6400

~~Nmap scan~~

TASK 5 - Nmap Host Discovery using ARP

1) How many hosts were found alive after scanning connection_IP/24?

* 1

Important Command

sudo nmap -PR -Sn CONNECTION_IP/24

-PR → ARP scan
-SN → Host discovery only

* ARP scanning is used when the target is on the same subnet. An ARP reply indicates that the host is active.

TASK 6 - Nmap Host Discovery using ICMP

Q1) what option is used for ICMP Timestamp discovery?

* -PP

Q2). what option is used for ICMP Address mask discovery?

* -PM

Q3). what option is used for ICMP Echo discovery?

* -PE

ICMP Echo:

nmap -PE -sn 10.200.6.0/24

ICMP Timestamp:

nmap -PP -sn 10.200.6.0/24

ICMP Address Mask:

nmap -PM -sn 10.200.6.0/24

TASK 7 - Nmap Host Discovery using TCP and UDP

Q1). which TCP Ping scan does not require a privileged account?

* TCP SYN Ping

Q2). which TCP Ping scan requires a privileged account?

* TCP ACK Ping

Q3). what option is required to run a TCP SYN Ping on the Telnet port?

* -PS23

TASK - 8 - Reverse DNS Lookup

Q1). which option is used to perform reverse DNS lookup for all hosts?

* -R

TASK - 9 - Summary

Q2). What is the purpose of -sn?

* It performs host discovery only without port scanning.

Q2). what is the purpose of -n?

* It disables DNS lookup.

Q3). what is the purpose of -R?

* It performs reverse DNS lookup for all hosts.

Q4). what option is used for ARP scan?

* -PR

Q5). what option is used for ICMP Echo scan?

* -PE

Q6). what option is used for ICMP Timestamp scan?

* -PP

Q7). what option is used for ICMP Address mask scan?

* -PM.

Q8). what option is used for TCP SYN Ping?

* -PS

Q9). what option is used for TCP ACK Ping?

* -PA

Q10). what option is used for UDP Ping?

* -PU.

Result: Thus, Nmap host discovery techniques such as ARP, ICMP, TCP & UDP were studied and used to identify live hosts in a network.